thermodynamics

Work done = area under pressure-volume diagram.

First law of thermodynamics applies to a **closed system**, which is one where matter cannot escape from or enter the system, but where an energy transfer into or out of the system is possible:

$$Q = W + \Delta U$$

- Q — the energy transferred into the system from the surroundings.
- W — the work done by the system on the surroundings.
- ΔU — the change in the internal energy of the system.

When a gas expands, work is always done **by** the gas (system); $W > 0$.
When a gas is compressed, work is always done **on** the gas (system); $W < 0$.
$W = 0$ implies that there is no change in volume; no work is done.

An **isolated system** is one that cannot interact with its surroundings. For example, an insulated box from which, or to, no energy can be transferred.

Isobaric $\implies$ constant pressure.
Isothermal $\implies$ constant temperature.
Isovolumetric / *Isochoric* $\implies$ constant volume.
Adiabatic $\implies$ no energy is transferred into or out of the system.

Note: Some papers may throw you off with *isochoric*. It is the same as *isovolumetric*.

W.r.t adiabatic change, $Q = 0$ so that $\Delta U = -W$.
Any external work done by the system **must** come from **internal energy**.
In other words, when work is done on the surroundings by the system, the **internal energy**, and hence the *temperature* of the system, must **decrease**.